

G32 INT4 payload

32 values share
one scale

HA-FBMS scale

32-bit coarse thermometer
and fine binary fields

Logical permutation

Characterization-weighted
software indices

Logical data flow only

No physical lane placement
No hardware fault-rate estimate
Equal width does not imply zero cost

One-step replay

Clean comparison and
selected logical faults

Packed kernel

Decode the scale and
reconstruct INT4 values

